

Interquartile Range (IQR) Outlier Calculation

Given Five Number Summary

Class	Minimum	Q1	Median	Q3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

Formula

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower Outlier Limit} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Outlier Limit} = Q3 + 1.5 \times \text{IQR}$$

Day Class

Calculation	Formula	Answer
IQR	$Q3 - Q1$	26.5
Lower Outlier	$Q1 - 1.5 \times \text{IQR}$	16.25
Upper Outlier	$Q3 + 1.5 \times \text{IQR}$	122.25

Minimum = 32

Maximum = 99

Result: No Lesser Outlier, No Greater Outlier

Night Class

Calculation	Formula	Answer
IQR	$Q3 - Q1$	11
Lower Outlier	$Q1 - 1.5 \times \text{IQR}$	61.5
Upper Outlier	$Q3 + 1.5 \times \text{IQR}$	105.5

Minimum = 25.5

Maximum = 98

Result: Lesser Outlier Exists (25.5), No Greater Outlier

Final Summary

Class	IQR	Lower Limit	Upper Limit	Lesser Outlier	Greater Outlier
Day	26.5	16.25	122.25	No	No
Night	11	61.5	105.5	Yes (25.5)	No